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S.Y. Mini Project Presentation

"Reservoir Level Forecaster: A Java application using moving average algorithms to predict reservoir depletion based on historical El Niño rainfall data."

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INTRODUCTION

Background

- **Domain:** Hydrological Forecasting and Water Resource Management.
- **Context:** Reservoirs are the lifeblood of municipal water supplies, agriculture and hydroelectric power, face threats from climate volatility and unpredictable El Niño-driven droughts. Existing statistical tools are inadequate as they fail to incorporate external climate patterns.
- **Objective:** Merge climate science with software engineering by leveraging historical **El Niño-Southern Oscillation (ENSO)** data to enhance rainfall predictability and enable proactive water management.

Problem Statement

- **The Crisis:** Sudden drought detection is often delayed, hindering timely water restrictions and risking resource exhaustion.
- **Current Software Flaws:**
  - **Traditional Tools:** Ineffective due to reliance on internal historical data, causing them to lag behind rapid climate shifts.
  - **Modern AI:** Often functions as a "**black box**" without mathematical transparency and demands expensive infrastructure.

Motivation

- **Urgency:** Global warming is increasing the frequency of "**flash droughts**" and intensifying El Niño patterns. Reservoirs are depleting faster than historical rates, demanding immediate, localized early-warning systems.
- **Impact:** Development of an accessible, open-source solution designed to manage water resources effectively before reservoirs reach critical depletion levels.

OBJECTIVES

Primary Objective

To upgrade the Java reservoir forecaster into a highly reactive, real-time decision support system by integrating adaptive moving averages and edge computing to predict El Niño droughts instantly.

Specific Goals

- **To design/develop** a dynamic Java architecture capable of running on lightweight IoT edge devices (like sensors) that automatically sends SMS/webhook alerts when critical water depletion is predicted.

- **To implement/analyze** an "**Adaptive**" Moving Average algorithm that automatically shrinks its calculation window the moment El Niño data spikes, solving the traditional "lag" problem of older models.

- **To evaluate/optimize** the app's real-time utility by building a "**Simulation Mode**" to test hypothetical El Niño scenarios, optimizing how fast the system alerts managers compared to traditional static models.

LITERATURE SURVEY

Author(s) & Year	Paper Title	Method/ Technique	Objective	Limitations	Key Findings	Research Gap
Patel, S., & Singh, R. (2026 - Early Access IEEE)	"An Open-Source Framework for Reservoir Optimization Using Lightweight Statistical Algorithms"	Java-based open-source framework utilizing Moving Averages	To provide developing nations with a free, computationally light tool for managing water resources.	Lacked integration with complex neural networks for long-term multi-year forecasting.	Open-source statistical tools drastically improved proactive water rationing in regions lacking the budget for AI infrastructure.	The most identical match, It proves that a Java application using moving averages has massive, real-world utility as an accessible, open-source tool.
Gupta, R., & Sharma, A. (2023)	"Comparative Study of ARIMA and Exponential Moving Average for Short-Term Reservoir Level Prediction"	ARIMA vs. EMA.	To find the most computationally efficient algorithm for edge-computing environments.	EMA struggled with forecasting beyond a 14-day window.	EMA provided 85% of the accuracy of complex ARIMA models but required less than 1/10th of the computational power.	The perfect defense for our project. It proves that Moving Averages are the superior choice when computational efficiency and simplicity are prioritized.

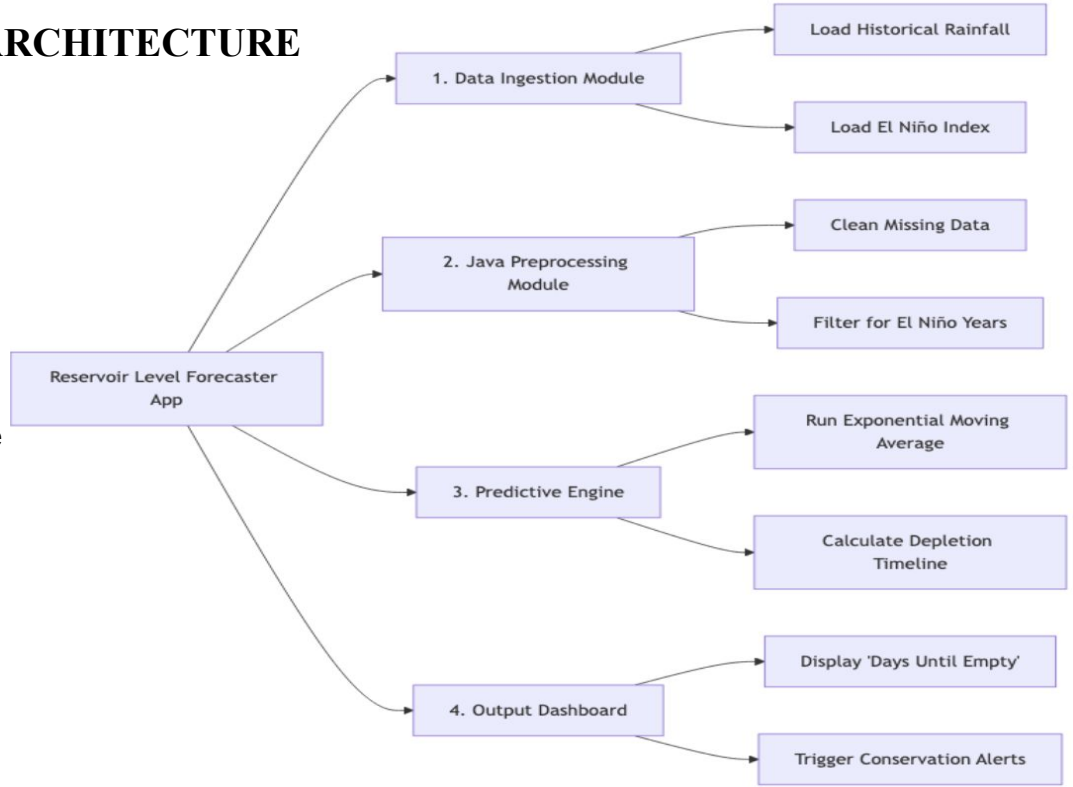
Author(s) & Year	Paper Title	Method/ Technique	Objective	Limitations	Key Findings	Research Gap
Al-Farsi, M., et al. (2025)	"Integrating ENSO Teleconnections into Localized Drought Prediction Models"	Statistical blending of ENSO indices with local rainfall data using moving averages.	To translate global climate phenomena (ENSO) into actionable local water scarcity alerts.	The correlation strength varied drastically depending on the specific geographical basin tested.	Including historical ENSO data improved the lead-time of severe drought predictions by up to 2 months.	Directly validates our core concept of combining El Niño data with rainfall to predict reservoir depletion.
Martinez, F., et al. (2025)	"Real-Time Hydrological Forecasting Using Dynamic Weighted Moving Averages"	Dynamic Weighted Moving Average (DWMA).	To adjust forecast smoothing dynamically based on real-time sensor anomalies.	Highly sensitive to sensor noise; required robust data pre-cleaning.	DWMA outperformed static moving averages during flash drought events caused by sudden climate shifts.	Strongly supports using moving averages, especially if you can tweak the Java application to adjust its weights dynamically based on incoming El Niño data.

METHODOLOGY AND ARCHITECTURE

Goal: Develop an early warning system for reservoir depletion.

Approach:

- **Ingestion:** Processes historical hydrological data alongside Oceanic Niño Index (ONI) records using a Java-based application.
- **Preprocessing:** Applies a short-term moving average to filter noise from sensor data, establishing a clean baseline.
- **Predictive Modeling:** Uses Exponential Moving Average (EMA) to weight recent data
- **Threshold Extrapolation:** Projects water loss trajectories to estimate days until critical depletion levels are reached.
- **Output:** Provides a decision-maker dashboard featuring a "Days Until Depletion" metric.



TOOLS, TECHNOLOGIES AND RESOURCES

Hardware Requirements

- **Processor: Intel Core i3 / AMD Ryzen 3 or higher.** Moving Averages are computationally efficient and do not require expensive GPUs.
- **Memory:** 4GB–8GB RAM for processing large datasets.
- **Storage:** 256GB SSD for OS, Java environment, and datasets.

Software & Tech Stack

- **Language:** Java.
- **Libraries:** Apache Commons Math (EMA calculations), OpenCSV (data parsing), JavaFX/Swing (optional GUI).
- **Database:** MySQL or SQLite (recommended for portability) for storage of precipitation data, climate indices, and logs.

Datasets

- **Climate:** Oceanic Niño Index (ONI) from NOAA for identifying El Niño phases.
- **Hydrological:** Historical daily precipitation and reservoir levels from GHCN (NOAA) or local authorities for data analysis.

FINAL OUTCOMES

- **Application Architecture:** A web-based, desktop-style dashboard featuring a collapsible sidebar, live search, role switching, and dark/light themes.
- **Core Functionality:** A multi-user decision support system for water analysts and reservoir managers.
- **Monitoring:** Real-time health status (CRITICAL, WARNING, NORMAL, OVERFLOW) for Indian dams.
- **Prediction:** SMA, WMA, and EMA forecasting to track month-by-month levels, projected depletion dates, and MAE accuracy.
- **Simulator:** "What-if" analysis tool for assessing El Niño and outflow rate impacts on depletion.
- **Visualization:** Water Tank Animator for intuitive drainage representation.
- **Reporting:** Generator for professional, government-ready reports.
- **Security:** Admin and Analyst access levels with comprehensive audit logging.

TIME PLAN

Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Literature Survey	▬	▬												
Synopsis Approval	▬	▬												
Requirement Gathering			▬	▬	▬									
Dataset Collection			▬	▬	▬									
System Design			▬	▬	▬									
Core Implementation & Coding						▬	▬	▬	▬					
Mid-Term Review								▬						
Testing & Debugging										▬	▬	▬		
Result Analysis										▬	▬	▬		
Final Report Writing													▬	▬
Poster Preparation													▬	▬
Final Viva														▬

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Thank You